

Jonathan J. Yang

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EDUCATION

The University of Texas at Austin

Austin, TX | May 2028

Bachelor of Science in Computer Science, **Turing Scholars Honors**

GPA: 3.96/4.00

Interests : Visual Arts, Swim

Coursework (*H=Honors*): *Algorithms (H)*, *Data Structures (H)*, *OS (H)*, *Computer Architecture (H)*, *AI (H)*, *ML (H)*, *Implementation of Modern Programming Languages (H)*, *Microprocessor Memory Systems (H)*, *Discrete Math (H)*

TECHNICAL SKILLS

Python, JavaScript/TypeScript, React, C, C++, Go, Rust, SQL, AWS, Linux, Docker, Git, PyTorch, CUDA, Verilog

PROFESSIONAL EXPERIENCE

Software Developer Intern @ Oracle Cloud Infrastructure (OCI)

Nashville, TN | May 2026 – Aug 2026

- Designed, built, validated, and deployed a full-stack monitoring application (React + OCI) for the Customer Compute Control Plane (CCCP) CI/CD team to monitor and diagnose flaky test failures in the build service.
- Designed streaming architecture to preview >200 TiB archive data with <1 GiB memory; user latency <1s.

Student Researcher @ UT LDOS Lab

Austin, TX | Nov 2025 – Present

- Extracted traces from Linux memory and scheduling module via eBPF probes to simulate thread-aware page cache prefetching.
- Transitioned to hypervisor-level context-aware disk-prefetching for concurrent guest workloads (e.g. GUPS+Zipfian).

Founding Member @ Apovo (Startup)

Austin, TX | Nov 2024 – Jul 2026

- Designed, scoped, and implemented requirements for a full-stack medical application (React Native + AWS). Targeted primary concerns of HIPAA compliance, secure identification + authentication, information correctness, usability, and operational cost.
- Engineered AI-powered voice navigation, hybrid matrix factorization + content-based search, and an auditable multi-agent annotation system for medical documents.
- Optimized search TTFB from 20s to 1s through caching, batch updates, model distillation, and progressive rendering.

Software Developer Intern @ Kent PLC

Dubai, UAE (Remote) | May 2025 – August 2025

- Developed assignment automation in ERP system to escalate tickets and leave requests to proper supervisors in Microsoft SQL database.

SELECTED PROJECTS

Programming Language Project (github.com/Jzhyang1/property-lang)

Sprint 2026

- Designed and implemented a language with built-in concurrent blocks, selective LLVM JIT-compilation, Python interoperability, and checked AI code-generation and testing/regeneration semantics.

Generative Visual Computing Projects

Fall 2025

- Implemented DCGAN, WGAN, and DDPM/DDIM image generation architectures for MNIST and CIFAR-10; motion generation with autoencoders; and nonparametric texture synthesis (pixel growth and patch-quilting).

High Throughput Kernel Lock-Free Scheduling Queue

Fall 2025

- Designed a custom lock-free queue optimized for a preemptive scheduler on a multicore operating system; achieves 75% throughput improvement over Michael-Scott queue for high contention cases.

Computer Architecture Class Project

Spring 2025

- Implemented pipelined RISC CPU (loads, stalls, arithmetic, branch prediction) in Verilog.
- Expanded project with a team to lower onto an FPGA; added peripherals, tilemap, and implemented Conway's game of Life and Snake on live system.

C++ Autograd Library for ML Training (github.com/Jzhyang1/cgr)

Summer 2025

- Built a C++23 autograd library with compile-checked tensor shapes, CUDA acceleration, common activations, and matmul optimizations (e.g. matrix slices, Strassen's algorithm)